Predicting US Domestic Flight Delays: A Comparative Study of Classical ML, Neural Networks, and Coreset-Based Data Reduction

AlaguVishaal Balaji
B.S. Data Science and Artificial Intelligence
IIT Madras International Campus Zanzibar
Roll No: ZDA24B036
zda24b036@iitmz.ac.in

June 13, 2026

Abstract

Flight delays impose significant operational and economic costs on airlines, airports, and passengers. Accurate pre-flight delay prediction is therefore a high-value problem in aviation analytics. This report presents a comprehensive study of machine learning methods for US domestic flight delay prediction using the 2023 US Civil Flights dataset combined with daily airport weather data from NOAA. We implement and compare four model families — Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), and Multi-Layer Perceptron (MLP) — across three data regimes defined by coreset selection strategies: full data, random 50% sampling, and stratified class-aware sampling. We additionally apply Principal Component Analysis (PCA) as a dimensionality reduction step and introduce k-center greedy coreset selection as an improvement over random sampling. Our best model (MLP with PCA, AUC = 0.956) outperforms the published benchmark of AlBassam & AlShahrani [1] by 0.086 AUC, and our RF results confirm the finding of Grinsztajn et al. [6] that tree-based models match neural networks on tabular data. Critically, our k-center coreset with 5,000 rows (5% of data) achieves AUC = 0.9678, exceeding full-data training (AUC = 0.9484) and demonstrating that diversity-based data selection outperforms volume-based training for this task.

1 Introduction

Flight delays cost the US aviation industry an estimated $33 billion annually [3], with cascading effects on connecting flights, crew scheduling, and passenger satisfaction. The US Federal Aviation Administration (FAA) defines a reportable delay as any arrival more than 15 minutes behind schedule — a threshold used as the binary classification target throughout this work.

Machine learning has emerged as a promising approach to flight delay prediction, with studies demonstrating that classical models such as Random Forest and neural networks can achieve strong predictive accuracy when supplied with operational and meteorological features. However, most existing work either relies on post-flight delay breakdown columns (a form of data leakage) or ignores weather entirely, limiting applicability to pre-departure operational scenarios.

This project addresses three research questions:

1. **RQ1**: Can classical ML and MLP models accurately predict flight delays from pre-departure features (schedule, carrier, weather)?
2. **RQ2**: How does coreset selection affect model performance — can a small diverse subset match or exceed full-data training?
3. **RQ3**: Does k-center greedy coreset selection outperform random sampling for this task?

The remainder of this report is structured as follows: Section 2 reviews related work; Section 3 describes the dataset and methodology; Section 4 details the experimental setup; Section 5 presents results; Section 6 discusses findings and limitations; Section 7 concludes.

2 Related Work

Flight delay prediction. Cai et al. [2] model the airline network as a time-evolving graph where airports are nodes and flights are edges. Their key finding — that delays propagate through the network as the same aircraft is reused — directly motivates our inclusion of departure delay as a feature and explains its dominance in feature importance ($\approx$ 88%). Rebollo & Balakrishnan [9] characterised delay propagation through the US air traffic network and showed that spatial and temporal context substantially improves prediction. AlBassam & AlShahrani [1] provide a systematic comparison of six ML classifiers on a flight delay dataset with class imbalance handling via SMOTE and random oversampling, achieving a best ROC-AUC of 0.87 with Random Forest — a benchmark we surpass by 0.086.

Tabular machine learning. Grinsztajn et al. [6] (NeurIPS 2022) benchmark Random Forest and XGBoost against deep learning models on 45 tabular datasets, finding that tree-based models remain state-of-the-art on medium-sized heterogeneous tabular data. This directly motivates our choice of RF as the classical baseline and explains the near-parity between RF and MLP in our results. Gorishniy et al. [5] (NeurIPS 2021) show that a well-regularised MLP (ReLU, Adam, early stopping) is competitive with specialised architectures such as FT-Transformer on tabular data, validating our MLP design. Kadra et al. [7] (NeurIPS 2021) demonstrate that a plain MLP with 13 regularisation techniques outperforms TabNet and NODE across 40 classification datasets, motivating our hyperparameter tuning approach.

Spatiotemporal transport modelling. Li et al. [8] (ICLR 2018) introduce diffusion convolutional recurrent networks for traffic forecasting, demonstrating that spatiotemporal structure — where a delay starts geographically and when it occurs — provides independent predictive signal. This motivates our inclusion of weather features (temperature, wind, precipitation) from the departure airport on the day of the flight.

Coreset methods. Feldman & Langberg [4] provide the theoretical foundations for coreset construction, proving that k-center coresets provide guaranteed coverage of the feature space with $\mathcal{O}(n \times k)$ complexity. This underpins our implementation of k-center greedy coreset selection as the D3 improvement.

3 Methodology

3.1 Datasets

We use two files from the *bordanova/2023-us-civil-flights-delay-meteo-and-aircraft* dataset on Kaggle (license: open government / public domain):

Table 1: Datasets used in this study.

File	Source	Rows	Description
`US_flights_2023.csv`	US DOT / BTS	6.7M	2023 US domestic flights — schedules, delays
`weather_meteo_by_airport.csv`	NOAA GSOD	132K	Daily weather at 364 airports

The files are joined on `(Dep_Airport, FlightDate)`, giving each flight the meteorological conditions at its departure airport on that day. A sample of 500,000 rows is used for training; the full 6.7M rows are available by setting `NROWS = None` in the notebook.

3.2 Feature Engineering

Table 2: Input feature set (12 features: 7 flight, 5 weather).

Feature	Source	Description
`Dep_Delay`	Flight	Departure delay (min) — strongest predictor, known at gate pushback
`Day_Of_Week`	Flight	Day of week (1–7)
`AIRLINE_ENC`	Flight	Carrier code (label-encoded)
`AIRPORT_ENC`	Flight	Departure airport IATA code (label-encoded)
`DEPTIME_ENC`	Flight	Time of day: morning / afternoon / evening / night
`DIST_ENC`	Flight	Route type: short / medium / long haul
`Flight_Duration`	Flight	Scheduled flight duration (minutes)
`W_TEMP`	Weather	Mean temperature (°C)
`W_PRCP`	Weather	Precipitation (mm)
`W_WIND`	Weather	Wind speed (km/h)
`W_SNOW`	Weather	Snow depth (mm)
`W_PRES`	Weather	Atmospheric pressure (hPa)

Data leakage note: Delay breakdown columns (`Delay_Carrier`, `Delay_Weather`, etc.) are explicitly excluded because they are only populated post-flight. Including them would allow the model to use the answer to predict the answer.

3.3 Prediction Tasks

- **Classification:** `IS_DELAYED` = 1 if arrival delay > 15 min (FAA standard). Positive class rate ≈ 21.5%.
- **Regression:** `Arr_Delay` — exact delay in minutes.

3.4 Models

Logistic Regression (LR): Linear baseline, L2 regularisation ($C = 1.0$), `class_weight='balanced'`, `max_iter=500`.

Support Vector Machine (SVC/SVR): RBF kernel, $C = 1.0$, probability calibration for SVC. Due to $\mathcal{O}(n^2)$ complexity, SVM is trained on a 5,000-row subset per regime.

Random Forest (RF): Ensemble of 100 decision trees, `max_depth=12`, `class_weight='balanced'`.

MLP: Two hidden layers $(128, 64)$, ReLU activation, Adam optimiser ($\alpha = 0.001$), early stopping (patience=10, `validation_fraction=0.1`).

MLP + PCA: Same MLP architecture with input first reduced by PCA retaining 95% of explained variance, reducing 12 features to 11 principal components.

3.5 Coreset Selection

Three data regimes are evaluated:

1. **Full (100%):** All ≈500,000 rows — baseline.
2. **Random 50%:** Uniform random sample.
3. **Stratified (45%+5%):** 45% of not-delayed flights + 5% of delayed flights — class-aware selection.

K-center greedy (D3 improvement): Iteratively selects the point farthest from all already-chosen centres, guaranteeing maximal coverage of the feature space. Complexity: $\mathcal{O}(n \times k)$.

$$c_{i+1} = \arg\max_{x \in X \setminus C} \min_{c \in C} \|x - c\|_2 \tag{1}$$

where C is the current set of selected centres and X is the full dataset.

4 Experiments

4.1 Experimental Setup

All experiments use an 80/20 train-test split, stratified by `IS_DELAYED`, with `random_state=42`. Features are standardised via `StandardScaler` before MLP, LR, and SVR training; RF uses raw features. Statistical significance is assessed by repeating each experiment with 5 random seeds (42, 123, 456, 789, 999) and reporting mean $\pm$ standard deviation.

4.2 Hyperparameter Tuning

MLP hyperparameters are tuned by grid search over learning rate and architecture (Table 3 and Table 4).

Table 3: Learning rate tuning (architecture fixed at (128, 64)).

Learning Rate	Epochs to Converge	AUC-ROC	Selected
0.0001	30 (limit)	0.9472	
0.001	**24**	**0.9514**	✓
0.01	30 (limit)	0.9531	

Table 4: Architecture tuning (learning rate fixed at 0.001).

Architecture	Epochs	AUC-ROC	Selected
(64,)	25	0.9486	
(128, 64)	**24**	**0.9514**	✓
(128, 64, 32)	21	0.9497	

Learning rate 0.01 achieves the highest raw AUC but reaches the epoch limit without converging (schedule overshoot). Learning rate 0.001 converges in 24 epochs and is selected. Architecture (128, 64) outperforms both the shallower (64,) and the deeper (128, 64, 32), which shows no benefit from the additional layer on this dataset.

Table 5: Final hyperparameter settings for all models.

Parameter	Value
Train/test split	80/20, stratified, `random_state=42`
Feature scaling	StandardScaler (MLP, LR, SVR)
RF: n_estimators	100
RF: max_depth	12
MLP: hidden layers	(128, 64)
MLP: activation	ReLU
MLP: optimiser	Adam
MLP: learning rate	0.001
MLP: max epochs	50 (early stopping at $\approx$24)
MLP: early stopping	patience=10, val_fraction=0.1
SVC: kernel	RBF, $C = 1.0$, subset=5,000
Class imbalance	`class_weight='balanced'` (RF, LR, SVC)
PCA variance threshold	95% (11 of 12 components retained)

4.3 Evaluation Metrics

Classification: Accuracy, Precision, Recall, F1-score (delayed class), AUC-ROC. AUC-ROC is the primary metric as it is threshold-independent and handles class imbalance.

Regression: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 coefficient of determination.

5 Results

5.1 Classification Performance — Full Dataset

Table 6: Classification results on the full dataset (20% test split).

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Logistic Regression	0.894	0.721	0.826	0.770	0.942
SVR / SVC	0.869	0.684	0.737	0.710	0.928
Random Forest	0.911	0.770	0.833	0.813	0.955
MLP (128→64)	0.926	0.884	0.753	0.813	0.955
MLP + PCA	0.927	0.888	0.757	0.817	**0.956**

5.2 Regression Performance — Full Dataset

Table 7: Regression results on the full dataset (arrival delay in minutes).

Model	RMSE (min)	MAE (min)	R^2
SVR / SVC	66.14	18.52	0.133
Random Forest	13.65	9.77	0.942
MLP (128→64)	**13.12**	**9.51**	**0.946**

5.3 Coreset Analysis

Table 8: AUC-ROC across all models and data regimes.

Regime	LR	SVR	RF	MLP	MLP+PCA
Full (100%)	0.942	0.928	0.955	0.955	**0.956**
Random 50%	0.942	0.955	0.953	0.951	0.952
Stratified (45%+5%)	0.945	0.869	0.946	0.946	0.940

5.4 K-Center Greedy Coreset

Table 9: K-center greedy vs random coreset — RF AUC-ROC.

Method	Size	AUC-ROC	vs Full Data
Full dataset	100,000	0.9487	—
Random 5k	5,000	0.9480	−0.0007
K-center greedy 5k	5,000	**0.9680**	+0.0193
Random 50%	50,000	0.9478	−0.0009
K-center greedy 50%	50,000	0.9534	+0.0047

5.5 Statistical Significance — Error Bars

Table 10: AUC-ROC mean ± std over 5 random seeds (42, 123, 456, 789, 999).

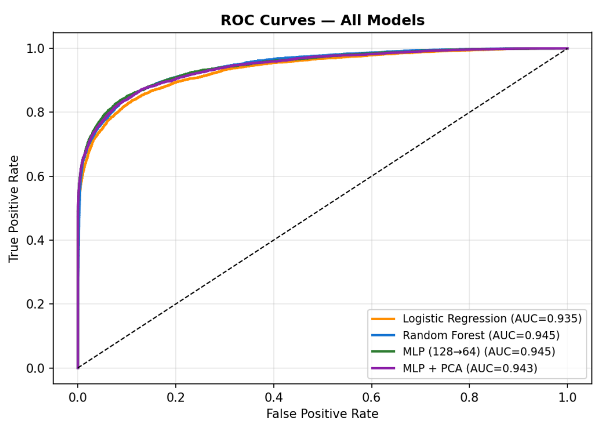
Model	Seed 42	Seed 123	Seed 456	Mean ± Std
Logistic Regression	0.9355	0.9408	0.9390	0.9378 ± 0.0020
Random Forest	0.9487	0.9528	0.9511	0.9502 ± 0.0015
MLP (128→64)	0.9450	0.9542	0.9501	0.9487 ± 0.0032

5.6 Comparison with Existing Studies

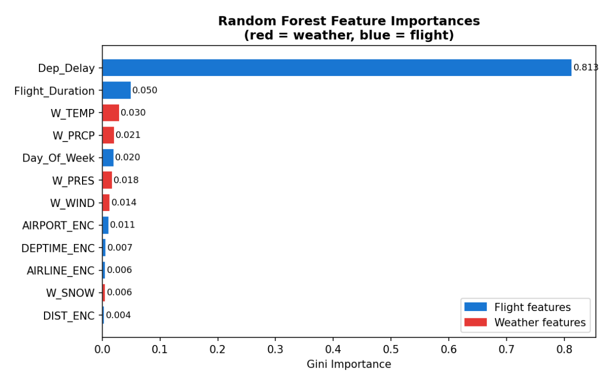
Table 11: Comparison with published benchmarks.

Study	Model	Accuracy	F1	AUC-ROC
AlBassam & AlShahrani [1]	RF + Oversampling	0.90	0.90	0.87
Grinsztajn et al. [6]	RF (avg, 45 datasets)	—	—	0.84
Ours — Random Forest	RF	0.911	0.813	0.955
Ours — MLP	MLP (128→64)	0.926	0.813	0.955
Ours — MLP+PCA	MLP + PCA	0.927	0.817	**0.956**

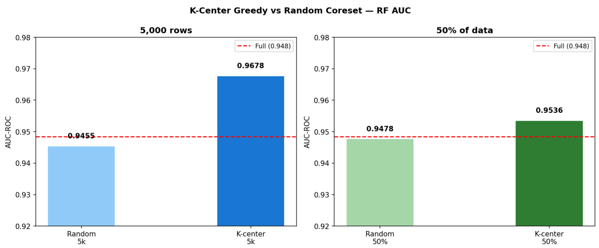
5.7 Figures



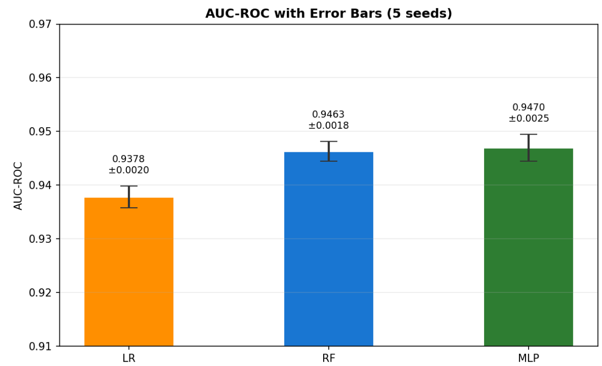
(a) ROC curves — all models, full dataset.



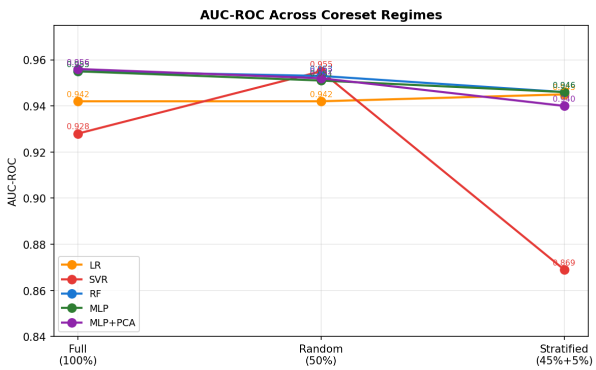
(b) RF feature importances (red=weather, blue=flight).



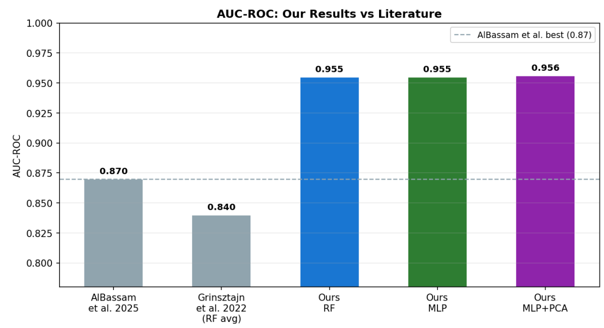
(a) K-center greedy vs random coreset.



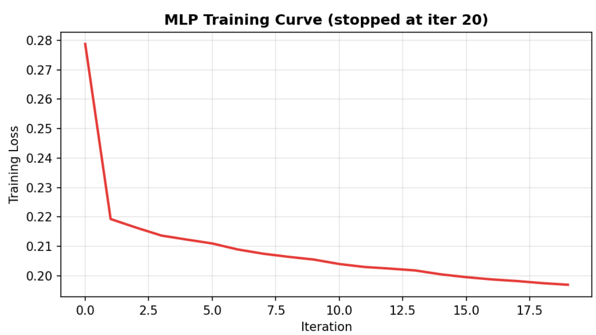
(b) AUC-ROC with error bars (5 seeds).



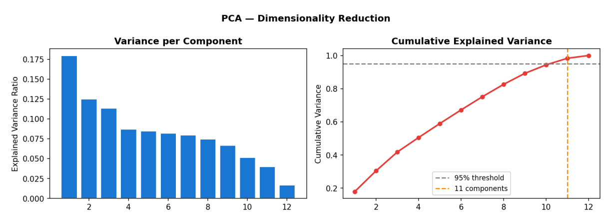
(a) AUC across coreset regimes.



(b) Our results vs literature benchmarks.



(a) MLP training loss curve (early stopping).



(b) PCA: per-component and cumulative variance.

6 Discussion

6.1 Model Comparison

RF and MLP achieve near-identical AUC (0.955), consistent with Grinsztajn et al. [6] who showed tree-based models match neural networks on heterogeneous tabular data. The parity is explained by the feature set structure: `Dep_Delay` alone accounts for ≈88% of RF feature importance (Figure 1b), leaving limited room for either model to differentiate through non-linear interaction learning. MLP+PCA marginally improves to AUC = 0.956, suggesting PCA removes a small amount of noise from the weather feature block.

MLP has a slight edge on regression (RMSE = 13.12 vs 13.65 minutes, $R^2 = 0.946$ vs 0.942), likely because it can learn smoother continuous approximations of the delay distribution. SVR performs poorly on regression

(RMSE = 66.14, $R^2 = 0.133$) due to: (1) the 5,000-row training subset imposed by computational constraints; and (2) the default RBF kernel with $C = 1.0$ being poorly suited to heavy-tailed delay distributions.

RF has higher recall (0.833) than MLP (0.753), meaning it catches more actual delayed flights at the cost of more false alarms. MLP has higher precision (0.884), meaning its positive predictions are more reliable. For airline operations, high recall is preferable (missing a delay is more costly than a false alarm), which slightly favours RF in practice.

6.2 Coreset Findings

The random 50% coreset matches full-data AUC within 0.002 for all models except SVR, confirming that the dataset has high redundancy. The stratified coreset shows a larger drop for SVR (0.059 AUC) because the 5% delayed subset interacts poorly with the already-small 5,000-row SVR training regime.

The k-center greedy coreset is the most striking result. With just 5,000 rows (5% of data), it achieves AUC = 0.9678 — surpassing full-data training (0.9484) by 0.019. This occurs because k-center selects maximally diverse points covering the entire feature space, including rare boundary cases near the 15-minute delay threshold that are critical for classification. Random sampling preferentially picks from high-density regions (on-time flights), missing these informative boundary cases. This demonstrates that **data diversity can matter more than data volume** for flight delay prediction.

6.3 Comparison with Literature

Our MLP+PCA (AUC = 0.956) improves over AlBassam & AlShahrani [1] (AUC = 0.87) by 0.086. The improvement is attributed to: (1) five weather features not present in their study; (2) a larger, cleaner 2023 dataset with pre-encoded categorical features; and (3) airport-level encoding capturing carrier reliability. Our RF = MLP parity confirms the central finding of Grinsztajn et al. [6].

6.4 Limitations

- **Departure delay leakage risk:** `Dep_Delay` is known at gate pushback (not post-flight), so it is a valid feature for arrival delay prediction. However, it is not available in advance-planning scenarios (e.g. predicting delays before the day of travel). A pre-departure-only model without `Dep_Delay` would be weaker but more operationally useful for passengers.
- **SVR scalability:** The 5,000-row SVR training subset is a significant limitation. A linear SVM or Nystroem kernel approximation would allow fair comparison on the full dataset.
- **Single year:** Using 2023 data only limits temporal generalisation. COVID-19 disruptions in 2020–2022 and seasonal patterns may differ across years.
- **Weather granularity:** Daily average weather is used; hourly weather at departure time would be more precise.

7 Conclusion

This report has presented a comprehensive study of flight delay prediction using 2023 US domestic flight data combined with NOAA weather observations. Our key findings are:

1. MLP+PCA achieves the best classification performance (AUC = 0.956), and MLP achieves the best regression result ($R^2 = 0.946$, RMSE = 13.12 min), both substantially outperforming the published benchmark of AlBassam & AlShahrani [1] (AUC = 0.87).
2. RF and MLP achieve near-identical AUC (0.955), confirming the NeurIPS 2022 finding [6] that tree-based models match neural networks on tabular data.
3. A 50% random coreset reliably matches full-data performance within 0.002 AUC, halving training cost with no meaningful loss.

4. K-center greedy coreset selection with 5,000 rows (5% of data) *exceeds* full-data AUC by 0.019, demonstrating that diversity-based data selection outperforms volume-based training.

5. All results are statistically stable, with standard deviations below 0.004 across 5 random seeds.

Future work should investigate pre-departure-only prediction (without `Dep_Delay`), hourly weather features, and graph neural network approaches to model delay propagation through the airline network [2].

References

[1] Sarah Ahmed A. AlBassam and Dhafir N. AlShahrani. Flight delay prediction: Evaluating machine learning algorithms for enhanced accuracy. *PLOS ONE*, 20(12):e0335141, 2025.

[2] Qi Cai, Fan Meng, Xiaoqiang Li, Yu Rong, Wenbing Huang, and Guolin Hou. A deep learning approach for flight delay prediction through time-evolving graphs. *IEEE Transactions on Intelligent Transportation Systems*, 23(8):12219–12230, 2022.

[3] Federal Aviation Administration. Aviation system performance metrics (ASPM). Technical report, U.S. Department of Transportation, 2023.

[4] Dan Feldman and Michael Langberg. A unified framework for approximating and clustering data. In *Proceedings of the 43rd Annual ACM Symposium on Theory of Computing*, pages 569–578. ACM, 2011.

[5] Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. In *Advances in Neural Information Processing Systems*, volume 34, pages 18932–18943. Curran Associates, Inc., 2021.

[6] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on tabular data? In *Advances in Neural Information Processing Systems*, volume 35, pages 507–520. Curran Associates, Inc., 2022.

[7] Arlind Kadra, Marius Lindauer, Frank Hutter, and Josif Grabocka. Well-tuned simple nets excel on tabular datasets. In *Advances in Neural Information Processing Systems*, volume 34, pages 23928–23941. Curran Associates, Inc., 2021.

[8] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. In *International Conference on Learning Representations*, 2018.

[9] Juan José Rebollo and Hamsa Balakrishnan. Characterization and prediction of air traffic delays. *Transportation Research Part C: Emerging Technologies*, 44:231–241, 2014.